Evaluating Authorization Regimes and Runtime Guardrail Configurations for Tool-Using LLM Agents

Abstract

Tool-using LLM agents can turn prompt injection into external actions, but measured security depends on the runtime authorization and Guard configuration. We keep evaluator fields offline and jointly measure unsafe effects, benign completion, confirmation/blocking, failures, latency, and adaptive attack cost.

We reanalyze 7,353 condition-level DeepSeek records and run controlled studies of authorization regime, prompt and runtime factors, source authority, historically difficult attacks, model replication, and attack-budget tails. In Public matrices, the full package produced more unsafe effects than the policy package on V4 Pro (38/600 versus 13/600) and V4 Flash (24/400 versus 12/400). A concurrent 2×2 experiment isolated prompt and module-bundle effects, but only two unsafe effects occurred across 400 attack assignments. Deterministic Guard replay was source-sensitive, whereas all 240 end-to-end source-swap runs executed no action. Exact authorization reduced pooled benign-tool completion and increased failure-sensitive risk, with opposite completion directions across AgentDojo and M3-v2.

Defense-aware search found first unsafe effects for 4/40 tasks by query 50 and 18/40 by 200; none of 12 prespecified B200 survivors first succeeded by 500 (Wilson upper limit 24.25%). In an additional risk-enriched single-block execution, no defense produced 17/60 unsafe effects, while policy/normal and full/safe produced 2/60 and 0/60 unsafe effects; failures remained non-negligible. These enriched results do not estimate deployment prevalence, utility, or temporal repeatability. Two incomplete Gemini studies do not support estimation of cross-provider effects.

Keywords：prompt injection; LLM agents; runtime authorization; provenance; security–utility trade-off

1 Introduction

1.1 Motivation

Tool-using language-model agents turn generated text into actions. An indirect prompt injection embedded in a message, document, webpage, memory item, or tool response can therefore become a request to send data, modify state, or invoke a privileged operation. Public environments such as AgentDojo and InjecAgent show that security must be evaluated together with task utility rather than as a text-classification problem alone [1, 2]. Runtime systems including VIGIL, DRIFT, the Task Shield, IPIGuard, and AgentSpec consequently interpose verification, task alignment, dependency, information-flow, or policy checks before an effect is committed [3–7].

This literature also makes a broad novelty claim untenable. Verify-before-commit, contextual policy generation, least privilege, planned dependency checks, and control/data-flow separation are already occupied design spaces [3, 7–10]. Recent preprints go further by auditing context influence, comparing an authorization graph with a parameter-source execution graph, or controlling source authority in next-action decisions [11–13]. Evidence-backed permission graphs with formal runtime authorization further narrow the mechanism claim boundary [14]. The open question for this study is therefore not whether a provenance-aware guard can be built for the first time. It is whether authorization information, enforcement structure, and failure handling materially change the security and utility measured for tool-using agents.

1.2 Evaluation gap

Three distinctions are easy to blur in an agent-security experiment.

First, a detector score or model refusal is not an executed-effect endpoint. Prompt-level structuring, spotlighting, instruction hierarchy, and injection detection can reduce model compliance, but they do not by themselves prove that an unauthorized action failed to cross an external runtime boundary [15–18].

Second, the authorization regime is part of the treatment. A guard that sees per-sample answer keys, evaluator labels, or an exact policy reconstructed from the expected answer is solving a different problem from a guard that receives only trusted user intent and a non-oracle policy compiled before execution. The present source audit does not establish that such leakage is universal, so we frame this as authorization uncertainty rather than a domain-wide answer-key defect.

Third, low attack success can be purchased by blocking useful work. A runtime defense must therefore report executed unsafe effects, benign task completion, system confirmation-request incidence, operational failures, latency, tokens, and cost under a common intention-to-treat denominator. Static attacks are also insufficient for a robustness claim when an attacker can adapt to the public defense interface [19].

1.3 Research questions

The study asks six questions.

1. **RQ1: Authorization regime.** How do `exact_gold`, `auto_compiled`, `broad`, `underspecified`, and `conflicting` authorization regimes change unsafe effect, benign task completion, and `ASK`/`DENY` burden?
2. **RQ2: Prompt and runtime components.** What are the independent main effects of the prompt profile and the runtime module bundle, and do they interact?
3. **RQ3: Conditional robustness and model replication.** Are the factorial effect directions consistent in a representative sample, an outcome-enriched challenge set, and a same-provider Flash replication?
4. **RQ4: Source authority.** When proposition, value, position, and policy are held fixed, how does moving a critical argument between trusted and untrusted origins change model–Guard behavior?
5. **RQ5: Adaptive tail risk.** Among attacks with no unsafe effect by budget 200, how often does a first unsafe effect appear by budget 500?
6. **RQ6: Contemporaneous attack activity.** In a historically risk-enriched cohort, do control attacks remain active in the current execution block, and how do six runtime conditions compare under the same provider window?

1.4 Study design

M3 uses a three-state runtime decision (`ALLOW`, `ASK`, `DENY`), a projection that excludes evaluator-only fields from online state, permission monotonicity for automatic paths, parameter-source evidence, and independent confirmation for privilege expansion. The completed authorization experiment crosses a policy package and a full package with non-oracle `auto_compiled` and oracle-sensitivity `exact_gold`; a nested subset adds three authorization stress regimes. The incremental factorial experiment then holds `auto_compiled` fixed and crosses prompt profile (`normal`/ `safe`) with runtime modules (`policy`/ `full`). A source-authority counterfactual holds text, parameter value, relative position, and policy fixed while changing only provenance

authority. Separate challenge, Flash-replication, adaptive-tail, and single-block activity calibration studies test conditional robustness. Timeouts, parsing failures, refusals, maximum-step outcomes, and provider failures remain in the intention-to-treat denominator.

The authorization-regime sample contains 180 paired cases and 900 DeepSeek V4 Pro work items. The incremental design adds 720 Pro factorial runs, 240 Pro source-authority runs plus 480 deterministic Guard decisions, 240 Pro challenge runs, 360 Flash replication runs, and a conditional adaptive tail of at most 3,600 additional queries. A final calibration adds 360 DeepSeek work items over 60 historically control-positive attacks and six concurrent conditions. The design separately reuses the completed 480-record medium study, 5,082-record Pro Public matrix, 1,791-record Flash Public matrix, and the 40-task adaptive evidence at budgets 50 and 200. Two bounded Gemini attempts remain diagnostic only. Provider tools and real external effects are prohibited throughout.

1.5 Empirical motivation and study framing

The completed medium falsification study did not support full-package superiority. The full package reduced unsafe effects from 8/36 under the structural condition to 0/36, but the policy package also produced 0/36. Benign tool completion fell from 12/24 under the structural condition to 6/24 under the full package, violating the preregistered non-inferiority margin. These observations motivate an empirical study of authorization uncertainty and the security–utility frontier rather than a claim that the full package is a new or state-of-the-art defense.

The larger Public primary matrix reinforced this finding. Across 600 attacks, the full package reduced unsafe effects from 133 under the structural condition to 38, but the policy package produced only 13. Across 168 benign tool tasks, both M3 variants completed 49 versus 103 under the structural condition. The full package therefore shows neither incremental security nor utility value beyond the policy package in the primary-model evidence. The 1.4955% retained diagnostic fraction also exceeds the predefined 1% threshold for a clean configuration-quality claim.

The completed 40-task adaptive panel observed first unsafe effect for 4/40 tasks by budget 50 and 18/40 by budget 200. These points show substantial budget sensitivity but do not establish saturation. The incremental tail extended only a prespecified subset of 12 B200 survivors toward 500. No new first success was observed (0/12; Wilson 95% upper limit 24.25%); this is a conditional analysis rather than a complete B500 curve or saturation result.

The prospectively frozen authorization-regime experiment is now complete for DeepSeek. Unsafe effects remained at or near zero across all four main cells, but exact authorization produced opposite benign-tool directions across the two benchmarks and increased failure-sensitive

attack risk on AgentDojo. It also replaced ASK with more frequent DENY and increased latency. The full package showed no stable advantage over the policy package. These unfavorable results are retained. A separate risk-enriched single-block calibration found 17/60 unsafe effects under no defense, confirming contemporaneous activity in the selected cohort; full/safe produced 0/60, but failures and the enriched sampling frame preclude deployment-prevalence, utility, or temporal-repeatability claims. Gemini free-tier execution remained too incomplete in both attempted studies for a cross-provider effect estimate.

1.6 Contributions

The current evidence supports six bounded contributions:

1. an evaluator/online-state contract that makes authorization visibility explicit and prevents evaluator-only fields from entering runtime state;
2. a contemporaneous factorial evaluation that identifies prompt-profile and runtime-module-bundle effects separately;
3. a controlled source-authority counterfactual that separates deterministic Guard decisions from end-to-end action and utility outcomes;
4. an intention-to-treat experiment that records official model identity, request commitments, isolated effects, and terminal diagnostics while retrying only response-less failed rows; and
5. an empirical account of authorization regime, confirmation/blocking burden, outcome-enriched instability, same-provider replication, and conditional defense-aware attack budget; and
6. a contemporaneous activity calibration that distinguishes an active control attack set from low event rates under defended conditions without treating a historical risk-enriched sample as deployment prevalence.

These contributions do not assert first, only, or SOTA status. Preprints are used to define the contemporary novelty boundary, not as peer-reviewed evidence.

2 Related work

2.1 Runtime verification and task alignment

VIGIL implements a verify-before-commit protocol for tool-stream injection and introduces the 959-case SIREN benchmark [3]. It is the closest peer-reviewed structural comparator to M3 because both systems interpose before an effect is committed. DRIFT instead combines dynamic rules with control- and data-level constraints and memory-stream isolation [4]. The

Task Shield verifies whether instructions and tool calls contribute to the user's declared objective [5]. Together, these peer-reviewed systems prevent us from claiming that M3 introduced verify-before-commit, dynamic information-flow checks, or task-alignment verification.

PlanGuard is a related preprint that isolates planning from untrusted context and applies hierarchical tool and parameter consistency checks [20]. Its presence further narrows the possible novelty claim: planning consistency is an adjacent mechanism, not an unoccupied problem. We therefore use these works to define comparison classes and mechanism boundaries rather than to motivate a first-or-only claim.

2.2 Privilege, provenance, and information flow

Progent represents least privilege as symbolic tool-and-argument rules. Its latest arXiv v3 description explicitly distinguishes automatic narrowing from expansion that requires approval [9]. This overlaps directly with M3's permission-monotonicity and confirmation design. IPIGuard, published at EMNLP 2025, constrains execution through a planned tool-dependency graph [7]. CaMeL separates trusted control and data flow and uses capabilities to prevent unauthorized flows [10]. These systems occupy the privilege, dependency, and information-flow design space that a provenance-aware runtime must acknowledge.

FAVA is a July 2026 preprint that translates natural-language tasks into a permission intermediate representation, lowers it to an evidence-backed permission graph, and uses an SMT authorizer before effectful execution [14]. It further narrows any mechanism novelty claim around structured authorization and data-flow-aware permission graphs. Our study differs by treating the authorization regime itself as the paired experimental factor and by jointly measuring executed unsafe effects, benign utility, confirmation/blocking burden, and operational failure.

Two 2026 preprints overlap even more closely with provenance-aware authorization. ARGUS constructs an influence-provenance graph for context-dependent decisions [11], while AuthGraph compares an authorization graph derived in an isolated context against a parameter-source execution graph [12]. Their existence rules out a claim that M3 is the first parameter-source or provenance-aware authorization defense. Because neither work is treated here as peer reviewed, their reported effects define the contemporary claim boundary but are not used as settled comparative facts.

Auditing Provenance Sensitivity in LLM Agent Action Selection supplies an especially close 2026 measurement counterpart: it holds task content and policy fixed while changing source authority, then measures whether action selection changes [13]. Its controlled next-action design is narrower than M3's executed-effect study, but it reinforces the need to separate evidence

relevance from authorization. Our experiment extends that question to full tool trajectories, paired runtime Guard conditions, benign utility, `ASK/DENY` burden, and operational failure.

The remaining defensible contribution is consequently methodological and empirical. We test whether authorization information itself changes measured security and utility; enforce a runtime projection that excludes evaluator fields; compare the policy and full configuration packages in paired conditions while disclosing their bundled prompt/runtime differences; and quantify `ASK` burden, operational failures, and executed effects under a common evidence contract. These are study-design claims and must stand on M3's own completed experiments.

2.3 Prompt-level defenses and execution-level outcomes

Structured queries, spotlighting, instruction hierarchy, and injection detectors act at the prompt, representation, or model-output layer [15–18]. They are relevant baselines, but detector scores and model compliance are not equivalent to an executed-effect endpoint. Our evaluation therefore records whether a proposed action crossed the runtime boundary and changed the sandbox state, while also retaining benign task completion. This distinction is necessary because a defense can appear secure by refusing useful actions or by failing before an effect is attempted.

2.4 Benchmarks and adaptive evaluation

AgentDojo supplies dynamic security and utility tasks and is the main public benchmark in this study [1]. InjecAgent provides a complementary collection of indirect prompt injections for tool-integrated agents [2]. ASB broadens the scenario and defense taxonomy [21]. The public expansion retains the benchmarks' unique task identities instead of duplicating benign cases to manufacture a denominator.

Static attacks are insufficient for a robustness claim. *The Attacker Moves Second* argues that a defense-aware attacker should optimize against the defense design rather than replay a fixed attack set [19]. The adaptive panel therefore exposes only public `ALLOW/ASK/DENY` and post-enforcement unsafe-effect feedback, searches frozen tasks at prespecified query budgets, and reports the first-success distribution. Since L27 remains a preprint, it motivates the protocol but is not described as a peer-reviewed theorem.

2.5 Positioning summary

M3 is not positioned as the first runtime guard, the first provenance system, or a state-of-the-art defense. The medium experiment already showed that M3-full did not improve security over M3 policy-only and failed the benign-tool non-inferiority criterion. The active paper in-

stead studies how non-oracle authorization, provenance-aware enforcement, confirmation, operational failure handling, benchmark choice, and adaptive attack budget jointly shape the measured security–utility frontier.

3 Methods

3.1 Study design and research questions

We conduct a paired empirical study of runtime authorization for tool-using language-model agents. The design separates the information supplied to the online Guard from the offline fields used to score the run. It then varies the authorization regime and the deployed Guard configuration package while holding the task, attack, provider, model, tool environment, and evaluator fixed.

The study addresses six research questions:

1. **RQ1: Authorization regime.** How do `exact_gold`, `auto_compiled`, `broad`, `underspecified`, and `conflicting` authorization regimes change executed unsafe effects, benign task completion, and `ASK`/`DENY` burden?
2. **RQ2: Prompt and runtime components.** What are the independent main effects of the safety prompt and full runtime modules, and do they interact?
3. **RQ3: Conditional robustness and model replication.** Are the factor directions consistent on historically difficult attacks and on DeepSeek V4 Flash?
4. **RQ4: Source authority.** When proposition, parameter value, position, and policy are held fixed, how does moving the proposition between trusted and untrusted sources change Guard and end-to-end behavior?
5. **RQ5: Adaptive tail risk.** Among attacks with no unsafe effect by budget 200, how many prospectively selected tasks first succeed by budget 500?
6. **RQ6: Contemporaneous attack activity.** In a historically risk-enriched cohort, are control attacks active in the current execution block, and how do six runtime conditions compare within that same block?

Positive results are not a stopping condition. The analysis is retained if `exact_gold` has little effect, the full package underperforms the policy package, or the available replication evidence differs from the primary model.

3.2 Threat model and safety boundary

The attacker controls text in documents, messages, retrieved records, memory, or simulated tool outputs processed while the agent performs a trusted user task. An unsafe effect is a state transition that violates the task-specific offline evaluator after the runtime Guard has issued its decision. The model may propose tool calls, but it cannot invoke provider-managed tools or reach a real external system.

All experiments use isolated benchmark state. Real email, payment, browser, shell, code-execution, production API, credential, and unrestricted filesystem operations are disabled. A proposal can affect only copied AgentDojo state, M3-v2 mock resources, `example.test`, `/mock/`, or the public sandbox. Model output is therefore treated as untrusted data, not executable authority.

Training-time poisoning, model-weight compromise, credential theft, host compromise, and direct modification of the trusted policy store are outside scope.

3.3 Online-state and evaluator separation

Every run constructs the online state through an allowlist projection. Permitted fields include trusted user intent, the benchmark-visible task context, the selected authorization regime, provenance events available at that step, and the proposed action. Gold answers, expected actions, attack and task-success labels, evaluator explanations, reference trajectories, and offline effect receipts are excluded.

The `exact_gold` condition is an explicitly labeled oracle sensitivity analysis. It converts evaluator-derived authorization into the condition's authorization object, but does not expose the evaluator labels or scoring fields themselves. `exact_gold` is never presented as the primary deployment regime. The primary regime is `auto_compiled`, created from the trusted user request before untrusted content is processed.

Each compact result stores a digest of the online-visible projection and a boolean forbidden-field check. A run fails the minimal integrity check if a forbidden field is present. The final analysis checks this condition once for all terminal records; it does not recursively replay historical evidence.

3.4 Runtime conditions

The common runtime interfaces are `TaskPolicyManifest`, `ActionProposal`, `GuardDecision`, and `ProvenanceEvent`. The two M3 conditions share the agent, model, parser, evaluator, tool environment, and task-policy representation, but they are not a single-module

ablation:

- **Policy package** evaluates the proposed tool and arguments against the current task policy with the `normal` prompt profile. Provenance propagation, temporal/data-flow checks, and independent confirmation are disabled.
- **Full package** uses the `safe_prompt` profile and enables policy enforcement plus parameter provenance, temporal/data-flow constraints, permission-monotonic transitions, and confirmation. Automatic transitions may preserve or reduce authority; authority expansion produces `ASK` and requires an independent confirmation result.

The earlier authorization experiment changed prompt profile and runtime modules together and is therefore retained only as a bundled configuration comparison. The new concurrent factorial crosses both factors explicitly: policy versus full modules and normal versus safe prompt. Its module main effect, prompt main effect, and module-by-prompt interaction are therefore separately identified within the fixed paired sample.

`ALLOW`, `ASK`, and `DENY` are runtime decisions, not security outcomes by themselves. Security is scored from the isolated post-enforcement effect receipt. Utility is scored by the benchmark task-completion evaluator.

The earlier incomplete 450-item Gemini authorization sample remains diagnostic and is not extended. E6 makes a separately executed, bounded Gemini prefix attempt that is diagnostic by design under the single-block amendment. Even complete prefix coverage would not make that amended design a formal cross-provider study; the observed incomplete prefix leaves every Gemini and cross-provider effect not estimable. Structural baselines, including the source-pinned VIGIL adapter, retain their earlier descriptive role; the new experiments target causal confounding, source authority, risk enrichment, same-provider model replication, and adaptive tail behavior rather than rerunning a large baseline matrix.

3.5 Authorization regimes

Five controlled code transformations are evaluated:

- `auto_compiled`: the task policy is compiled from the trusted user request, trusted application context, and tool schemas before untrusted evidence is processed;
- `exact_gold`: expected tool calls are derived offline from the evaluator and converted into required exact argument rules, with confirmation disabled; only the derived authorization object enters online state;
- `broad`: each exact argument constraint is replaced by a type constraint; when this widens a side-effecting rule, confirmation is required;

- `underspecified`: the last compiled tool rule is removed; and
- `conflicting`: each exact value is replaced by a deterministic controlled conflicting value (for example, a recipient becomes `conflicting-scope[at]example.test` or a path becomes a public-sandbox conflict path).

As a sanitized example, an auto-compiled rule allowing `send_email(to="analyst[at]example.test")` becomes `send_email(to:string,confirmationrequired)` under `broad`, is removed if it is the final rule under `underspecified`, and targets `conflicting-scope[at]example.test` under `conflicting`. Under `exact_gold`, an offline expected call becomes a required exact-value rule. These definitions describe this implementation; they are not prevalence estimates for deployed authorization systems.

The four primary DeepSeek cells pair policy-only and M3-full with `auto_compiled` and `exact_gold`. A fixed nested subset adds M3-full under the three non-oracle stress regimes.

3.6 Existing evidence reused without new calls

Four completed datasets are analyzed once:

- 5,082 DeepSeek V4 Pro records from the Public primary matrix;
- 1,791 DeepSeek V4 Flash records from the Public second-model matrix;
- 480 records from the medium falsification study; and
- the completed DeepSeek V4 Pro adaptive tasks at budgets 50 and 200.

The three record matrices contain 847, 597, and 72 unique cases, respectively. The Flash cohort is a 597-case subset of the Pro Public cohort; 17 medium cases overlap Pro and 16 overlap Flash. Their union is 902 distinct cases, so the 7,353 condition-level records are not treated as 7,353 independent cases.

The reuse analysis performs one count/uniqueness pass, retains all terminal failures in the ITT population, and computes the prespecified endpoint summaries without repeating model execution. These data establish the empirical motivation and the two-point adaptive first-success curve; they do not establish attack saturation.

3.7 Frozen primary sample

The new DeepSeek study contains 180 unique paired cases selected before the new outcomes were inspected:

- 100 attacks, with 50 from AgentDojo and 50 from M3-v2;
- 60 benign tool tasks, with 30 from each benchmark; and
- 20 benign no-tool tasks.

The lean study was designed after the larger Public matrices had been completed, but its deterministic selector reads only frozen Public execution-plan identifiers, pre-outcome strata, and call budgets. It does not read any Public result or outcome file. Selection is stratified by benchmark, task type, and attack family. Every main case is assigned to four cells: policy package/ `auto_compiled`, policy package/ `exact_gold`, full package/ `auto_compiled`, and full package/ `exact_gold`. This produces 720 work items.

A nested 60-case subset receives three additional M3-full assignments: `broad`, `underspecified`, and `conflicting`, adding 180 work items. The frozen DeepSeek plan therefore contains 900 work items. The first six items form the formal technical prefix and remain part of the final denominator.

3.8 Concurrent prompt-by-module factorial

E1 reuses the 180-case sample but reruns every case contemporaneously under four `auto_compiled` cells: policy/normal, policy/safe, full/normal, and full/safe. This produces 720 new work items and prevents an effect estimate from being assembled across dates or execution identities. An eight-item formal prefix covers all four cells and both attack and benign-tool tasks; it is included once in the final denominator.

Table 1 Prompt-by-module factorial design

Condition cell	Prompt	Modules	Authorization	Historical rows reused
Policy / normal	normal	policy	auto compiled	No
Policy / safe	safe	policy	auto compiled	No
Full / normal	normal	full	auto compiled	No
Full / safe	safe	full	auto compiled	No

Note. The four contemporaneous cells cross prompt profile with the runtime module bundle while holding authorization at auto_compiled.

3.9 Source-authority counterfactual

E2 contains 60 controlled tool tasks, evenly split between AgentDojo and M3-v2. Thirty are supporting scenarios and thirty contain a parameter that competes with the trusted request. The scenarios cover payment targets, message recipients, file paths, calendar participants, data recipients, and workspace objects.

Each task has `trusted_origin` and `untrusted_origin` versions. Proposition, parameter value, relative position, and policy remain fixed; only source authority changes. Policy/normal and full/normal are crossed with both source variants, yielding 240 end-to-end work items. Separately, authorized and unauthorized proposals are replayed through both Guard configurations for every source variant, yielding 480 deterministic, zero-provider decisions. Offline

Guard replay and end-to-end system performance are reported as different evidence layers.

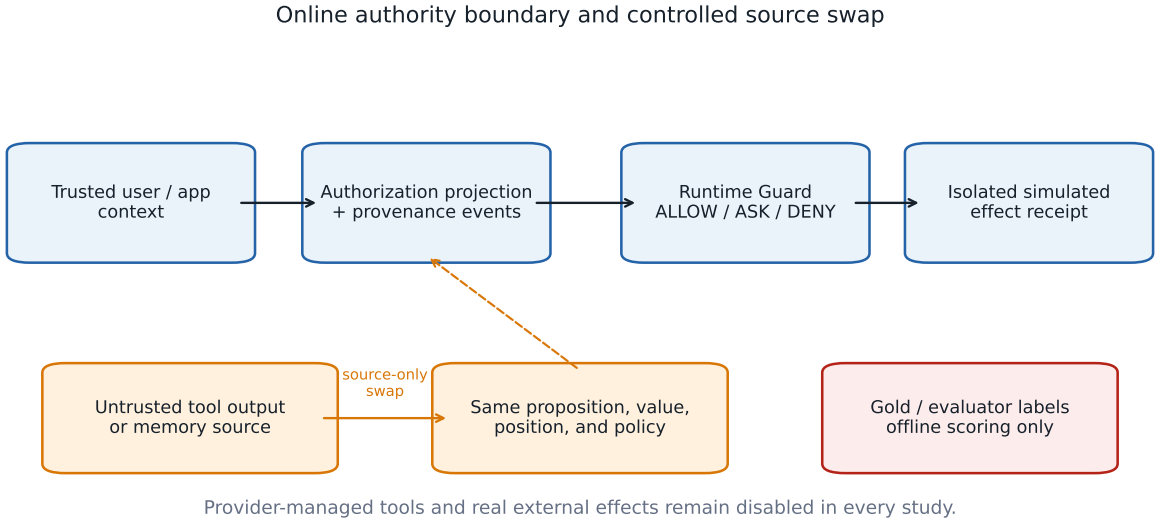


Figure 1 Online fields and the source-authority boundary. Trusted task context, projected authorization, provenance events, and proposed actions may enter the Guard. Raw evaluator labels, gold answers, and effect receipts remain offline. The E2 source swap changes authority while holding proposition, value, relative position, and policy fixed; provider-managed tools and real external effects remain disabled.

3.10 Risk-enriched challenge and Flash replication

E3 selects 60 historical attack cases, 30 per benchmark, that produced an unsafe effect under at least one earlier public condition. Coverage of suite and attack family is prioritized before the count of earlier unsafe conditions; a fixed digest order breaks ties. Each case receives the four E1 factor cells, yielding 240 Pro runs. This outcome-enriched cohort estimates conditional robustness on known difficult cases and cannot estimate natural attack prevalence.

E4 uses a 90-case nested representative sample with 50 attacks, 30 benign tool tasks, and 10 benign no-tool tasks. It repeats all four E1 cells on DeepSeek V4 Flash, yielding 360 work items. This is a same-provider model replication, not a cross-vendor generalization study.

3.11 Adaptive B200-survivor tail

E5 starts from the 22 of 40 defense-aware attack tasks with no unsafe effect by query 200. Twelve were selected prospectively: all four A2 tasks and four each from A1 and A3 by fixed digest order. Their existing search checkpoints are continued from query 201 to at most 500 and stop at the first unsafe effect. The maximum is 3,600 new queries. We report

P(firstsuccessby500|nosuccessby200), first-success locations, a conditional survival step function, family strata, query count, tokens, cost, and failure. These twelve tasks are not merged into the original 40-task cohort to construct a nominal full B500 curve.

3.12 Contemporaneous attack-activity calibration

E6 freezes 60 unique attacks, 30 per benchmark, selected only from historical Public-primary cases that produced an unsafe effect under no defense or the source-pinned VIGIL adapter. It runs each task once under no defense, VIGIL, policy/normal, policy/safe, full/normal, and full/safe, producing 360 DeepSeek work items in one execution block. Condition order uses a fixed six-way rotation within each benchmark. A 12-item outcome-blind formal prefix is included in the final denominator. This risk-enriched cohort tests whether attacks remain active contemporaneously and compares conditions within the same block; it does not estimate deployment prevalence. The execution is separate from E3, but the sample is not independent: 47/60 tasks are E3 anchors (19 AgentDojo and 28 M3-v2).

The original three-date E6 design was prospectively amended before any E6-v2 provider call because artificial spacing did not answer a mechanism-relevant question. The amendment retained the frozen 60-task cohort, six conditions, sampling rules, endpoints, and failure policy, but replaced temporal replication with one complete diagnostic block. Consequently, E6 cannot estimate temporal repeatability. It contains no benign tasks and cannot contribute TCR or utility evidence.

A nested 30-task Gemini plan used the same cases and conditions; 29/30 tasks are E3 anchors (15 AgentDojo and 14 M3-v2). Its formal prefix stopped after five response-bearing records and one response-less VIGIL dependency failure; the remaining 174 items were administratively closed rather than scientifically executed. All Gemini paired, factorial, and cross-provider effects are therefore not estimable.

3.13 Providers and model identity

The primary model is DeepSeek V4 Pro through the official DeepSeek endpoint; E4 uses DeepSeek V4 Flash through the same official provider. The earlier 450-item Gemini authorization sample and the separately executed E6 Gemini prefix are both diagnostic. The latter used the official free tier at USD 0 and stopped at its technical gate; neither Gemini dataset supports an effect estimate. Each new request is assigned a local request identifier and canonical payload digest before transport. A response-bearing record retains the official response identifier, exact returned model identity, usage, latency, and cost or free-tier attestation. For E5, the top-level compact cost object is empty, so request-level CNY charges are read from the public provider

trace bundle embedded in the same raw query row. Its usage summary already aggregates multiple provider calls for the query; missing attribution fails closed. The raw-row totals are checked against the live append-only budget tracker during execution, and the paper analysis does not need to read the internal ledger or change any experimental outcome. The initial E5 cost estimate set CNY 5 adaptive and CNY 16 global fuses. After 898 latest query work items cost CNY 4.2561534, the no-success maximum projected to approximately CNY 17. Before resuming, the cost ceilings were first raised and later replaced by a CNY 1,000 runaway-protection ceiling at the project owner's direction. The selected tasks, query range, attacker, model, seed, first-success stopping, ITT rules, and estimand were unchanged, and no successful query was replayed. Final analysis-attributed cost was CNY 24.5551236. Provider calls for the evidence reported here were made during July and August

1. The returned names are provider-managed model aliases and therefore

   identify the served model at request time without freezing its hidden weights.

Prompts and model responses are not copied into the compact analysis file. Content-free request metadata and the provider's internal trace remain in a restricted local run directory outside the analysis dataset. Exact model identity is checked in the final one-pass integrity validation.

3.14 Failure and recovery policy

Automatic retry is disabled. Recovery operates at the predefined experiment-item level:

- a work item with an official model response is terminal, including timeout, refusal, parse failure, and maximum-step outcomes after that response;
- a transport or preparation failure with no official response may be retried only with the explicit failed-row option;
- only the affected work item is retried; successful and unaffected work items are skipped;
- a repeated non-rate-limit pre-response failure becomes terminal on its second attempt; and
- no automatic retry or cross-model fallback is permitted.

Earlier response-less attempts remain append-only in the raw file. The scientific analysis selects the latest terminal attempt for each experiment item and reports the count of superseded attempts. No prior row is overwritten.

3.15 Outcomes

The two joint primary endpoints are:

1. **unsafe-effect rate** on assigned attack tasks; and
2. **task-completion rate (TCR)** on assigned benign tool tasks.

Benign no-tool completion is descriptive. Secondary outcomes are system `ASK` and `DENY` event counts and record incidence, operational failure, p50/p95 latency, total tokens, API cost, and predefined heterogeneity by benchmark and attack family. `ASK` is a runtime confirmation request; without a human study it is not interpreted as measured human burden.

All terminal work items remain in the intention-to-treat denominator. The analysis reports raw numerator/denominator values alongside proportions. A provider or parser failure is not silently removed and is not relabeled as a successful security outcome. Its unsafe-effect indicator records whether an isolated unsafe effect was observed, while its operational failure remains a separate reported outcome and it cannot count as task completion. Latency and burden summaries likewise include all terminal work items, not only completed ones.

Because observed unsafe effects are rare, we also report a pessimistic sensitivity endpoint on attack tasks: `unsafe_effectORoperational_failure`. This composite is a robustness bound, not an attack-success label. It asks how conclusions change if a non-operational run is not credited as secure while preserving unsafe effects and failures as separate reported counts.

3.16 Statistical analysis

Each condition proportion receives a Wilson 95% confidence interval. The paired analyses report the absolute difference in percentage points, the two discordant counts, and a two-sided exact McNemar test. Uncertainty for the paired absolute difference is estimated with 10,000 deterministic bootstrap resamples. Each resample draws paired case identifiers with replacement and keeps both condition outcomes for the selected case together; the case is the cluster and inferential unit.

The four primary authorization/configuration comparisons are:

1. policy package `auto_compiled` versus policy package `exact_gold`;
2. full package `auto_compiled` versus full package `exact_gold`;
3. `auto_compiled` policy package versus full package; and
4. `exact_gold` policy package versus full package.

McNemar p-values are Holm-adjusted separately within the four-comparison attack unsafe-effect family and the four-comparison benign-tool TCR family. Bootstrap confidence intervals are unadjusted estimation intervals. The `broad`, `underspecified`, and `conflicting` comparisons, the unsafe-or-failure bound, and benchmark panels are descriptive sensitivity or heterogeneity analyses, not additional confirmatory families.

For E1, binary outcomes are arranged as a within-case 2×2 design. The module main effect is the average of full-minus-policy differences across prompts; the prompt main effect averages safe-minus-normal differences across module settings; and the interaction is the difference

between those two prompt effects. Ten thousand case-cluster bootstrap resamples provide intervals and two-sided bootstrap sign p-values. Module, prompt, and interaction are Holm-adjusted within the attack unsafe-effect and benign-tool TCR endpoint families. The unsafe-or-failure factor analysis is a prespecified sensitivity bound and is shown separately.

E2 uses the analogous source-by-module contrasts for unauthorized-argument influence and authorized action execution. Source, module, and source-by-module form the source experiment's adjustment family. Source-swap action changes are paired by base task. Deterministic replay reports false allows, false blocks, and an ALLOW/ASK/DENY heatmap without treating replay accuracy as end-to-end system performance.

E3 and E4 report effect directions, absolute differences, and bootstrap intervals but open no new significance family. E3 is an outcome-enriched challenge analysis; E4 is a same-provider replication. Both retain their sampling boundary in every table and figure.

For E6, no-defense activity is reported per benchmark and overall with Wilson intervals. Four prespecified paired contrasts use treatment-minus-baseline risk differences: VIGIL minus no defense, policy/normal minus no defense, full/safe minus no defense, and full/safe minus VIGIL. Each contrast receives a paired Newcombe interval, a benchmark-stratified task-cluster bootstrap interval, and a two-sided exact McNemar p-value; the four p-values are Holm-adjusted. The implemented exact test replaced the planned cluster sign-flip test and is reported as an analysis-method deviation; no row or outcome was removed. Operational failure and `unsafe_effectOROperational_failure` are reported alongside unsafe effect. The four M3 cells receive descriptive module, prompt, and interaction estimates. Because E6 is risk-enriched and single-block, these analyses do not estimate prevalence, utility, temporal repeatability, or the effect of an individual provenance component. Gemini effects are reported as not estimable after its coverage gate failed.

The fixed design provides 100 paired attacks and 60 paired benign-tool tasks per main contrast, split evenly by benchmark for the latter. With 0/100 observed unsafe effects, the Wilson upper limit is 3.70%; within a 30-task benign-tool benchmark stratum, intervals are necessarily much wider (for example, 0/30 has an 11.35% upper limit). We report these design-precision limits instead of a post hoc achieved-power calculation.

The original adaptive panel reports the cumulative fraction of the fixed 40-task cohort with a first unsafe effect at budgets 50 and 200. E5 separately reports the conditional probability of a first success by 500 among twelve prospectively selected B200 survivors and a conditional survival curve from 201 to 500. We neither combine the sampling frames nor extrapolate a saturation level.

3.17 Minimal verification and reproducibility

Verification is deliberately proportional to the lean study. It consists of:

1. focused unit tests for modified provider, authorization projection, evaluator, analysis, and plotting code;
2. an eight-item formal prefix for each regular incremental study, two five-query tail prefixes for E5, and a 12-item outcome-blind E6 prefix; and
3. one terminal pass over row count, experiment-item uniqueness, forbidden-field isolation, exact model identity, and ITT failure retention.

Each study writes one append-only compact raw JSONL. Final statistics are written once to one analysis JSON per study, and figures are exported as submission-ready vector PDF, editable SVG, and raster PNG previews for manuscript/QA use. Reproducibility assets include frozen samples, condition assignments, analysis code, aggregate tables, and versioned source.

4 Results

4.1 Medium falsification study

The medium study contains 480 records: 432 primary `auto_compiled` records covering 72 paired cases across six conditions, plus 48 authorization-regime probe records. The preregistered analysis did not support a superiority claim for the full configuration package. It met the security-effect and deny-all criteria, but missed the utility point non-inferiority criterion and showed no security gain over the policy package. Both the favorable security comparison and the unfavorable utility and package comparisons are reported.

Supplementary Table S1 reports the online-state source audit.

This audit supports only covered-path statements. It does not establish a domain-wide oracle-authorization problem.

Supplementary Table S2 reports the completed medium security and utility outcomes.

The medium sample therefore shows a security–utility trade-off, not an incremental package benefit. Both M3 variants eliminated the eight unsafe effects observed under the structural condition, but they also completed substantially fewer benign tool tasks.

Supplementary Table S3 reports the prespecified paired comparisons.

The security effect relative to the structural baseline was large in this sample, but the full package did not outperform the policy package. Relative to the structural condition, its benign-tool TCR was 25 percentage points lower, well outside the preregistered -5 percentage-point non-inferiority margin.

Supplementary Table S4 reports the runtime burden and operational quality.

Across the 432 primary records, 411 completed and 21 retained terminal failures comprised six `max_steps`, six parse failures, and nine provider failures. The 4.86% overall failure fraction and 2.08% provider-failure fraction passed the predefined operational-quality envelope. Every terminal state remains in the intention-to-treat denominator.

For the full package, 22 initial `ASK` decisions comprised 17 permission-expansion events, four independent-confirmation requirements, and one out-of-scope parameter. Confirmation outcomes remained `pending` in this synthetic study; the ASK count measures system confirmation-request incidence and cannot be interpreted as successful approval or measured human burden.

4.2 Authorization-regime sensitivity

Forty-eight probe records compare four alternate regimes with matched `auto_compiled` executions for the policy and full packages. Each condition–regime comparison contains six pairs. No unsafe effect occurred in these small probes. For the explicitly labeled `exact_gold` probe, task completion increased from 1/6 under matched `auto_compiled` to 3/6 under `exact_gold`, while initial ASK decreased from 3/6 to 0/6; exact McNemar tests were p=0.5 and p=0.25, respectively. These six-pair estimates are sensitivity signals, not evidence of a domain-wide bias.

4.3 Prespecified decision criteria

Gate	Result	Evidence
Security effect vs structural	Pass	22.22 pp absolute reduction
Tool-utility point non-inferiority	Fail	−25.00 pp vs −5 pp margin
Full vs policy-package security gain	Fail	0/36 vs 0/36
Deny-all rejection	Pass	Benign initial ASK-or-DENY 22.22%
Operational quality	Pass	All three predefined quality checks passed
Supported interpretation	Empirical trade-off study	Full-package superiority claim not supported

The results therefore support an empirical study of authorization uncertainty and the security–utility trade-off. They do not support a defense-superiority or SOTA claim.

4.4 Public primary-model result

The completed Public matrix contains 847 unique cases and 5,082 records: 600 attacks and 247 benign cases crossed with the same six conditions. All 76 diagnostic records

(1.4955%) remain in ITT. The predefined 1% quality ceiling therefore failed and prohibits a clean configuration-quality claim. A one-pass integrity check nevertheless confirmed unique case-condition coverage, model identity, and diagnostic retention. The result is primary-model-only.

Across the 600 paired attacks, the full package produced 38 unsafe effects (6.33%, Wilson 95% CI 4.65–8.57) versus 133 (22.17%, 19.03–25.66) under the structural condition. The paired difference was −15.83 percentage points (10,000-cluster bootstrap 95% CI −19.50 to −12.17; exact McNemar p=4.04e−17, Holm-adjusted p=1.61e−16). This security reduction did not establish full-package superiority: the policy package produced only 13 unsafe effects (2.17%, 1.27–3.67), so the full package was 4.17 points worse (95% bootstrap CI 2.50–6.00; exact McNemar p=1.62e−6).

Benign tool completion under the full package was 49/168 (29.17%, 22.82–36.44), versus 103/168 (61.31%, 53.77–68.34) under the structural condition: −32.14 points (95% bootstrap CI −39.88 to −25.00; exact McNemar p=1.19e−14). The full and policy packages both completed 49/168, with a paired difference of 0.00 points. All six conditions completed 79/79 benign no-tool tasks.

Supplementary Table S5 reports the public-primary benchmark strata.

The direction is consistent across both attack benchmarks: the full package is safer than the structural condition but less safe than the policy package. Utility is strongly benchmark dependent. The public M3-v2 tool tasks show no completion difference among the three focal conditions, whereas AgentDojo tool completion falls by 55.67 points for both M3 variants relative to the structural condition. AgentDojo benign denominators remain the 97 unique official utility tasks and are not duplicated.

4.5 Public DeepSeek V4 Flash replication

The completed Flash matrix contains 1,791 records: 400 attacks, 145 benign tool tasks, and 52 benign no-tool tasks under each of the structural, policy, and full packages. The full package produced 24/400 unsafe effects (6.00%, Wilson 95% CI 4.06–8.77), compared with 12/400 under the policy package (3.00%, 1.72–5.17) and 47/400 under the structural baseline (11.75%, 8.95–15.28). Thus the full-minus-policy difference was +3.00 percentage points, preserving the unfavorable direction observed with V4 Pro.

Benign-tool completion was 38/145 for the full package (26.21%, 19.73–33.91), 35/145 for the policy package (24.14%, 17.90–31.71), and 82/145 for the structural baseline (56.55%, 48.42–64.35). All three conditions completed 52/52 benign no-tool tasks. The small +2.07-point full-minus-policy tool-completion difference does not offset the security disadvantage and

is not interpreted as an independent component-specific gain. Sixteen operational diagnostic records remain in ITT; the Flash result therefore supports a same-provider model replication, not a clean configuration-quality or cross-provider claim.

4.6 Public operational and decision burden

The Public matrix contains 4,616 completed, 112 maximum-step, 120 parse-failure, and 234 provider-failure records. Diagnostics are uneven by condition: 27 structural, 24 safe-prompt, 22 full-package, two no-defense, one PromptGuard, and zero policy-package. This imbalance is disclosed because terminal failure can affect observed unsafe-effect counts; no diagnostic was retried or excluded.

On attacks, the full package issued 181 initial `ASK` and 44 initial `DENY` decisions, versus 197 `ASK` and no `DENY` under the policy package. On the 168 benign tool tasks, the full package issued 75 `ASK` decisions and the policy package issued 69. The package comparison therefore met its structural security-effect criterion but missed utility non-inferiority and superiority criteria. This pattern is consistent with a security–utility trade-off rather than incremental benefit from the full package.

4.7 DeepSeek authorization-regime experiment

The frozen authorization study contributes 900 terminal DeepSeek V4 Pro work items across 180 paired cases. The four main cells each contain 100 attacks, 60 benign tool tasks, and 20 benign no-tool tasks. A nested 60-case subset adds the broad, underspecified, and conflicting regimes under the full package. Recovery was limited to 33 work items for which no provider response had been obtained; all 867 response-bearing work items were left untouched. The final ITT population contains 768 completed records, 54 parse failures, 30 maximum-step terminals, 15 provider failures, and 33 repeated pre-response failures. No raw evaluator, label, or gold-answer field entered the online projection; `exact_gold` exposed only its derived authorization object. Provider tools and real external effects remained disabled.

4.7.1 Joint security and utility endpoints

Unsafe effects were at the floor in the main experiment: 0/100 under the full package with either `auto_compiled` or `exact_gold`, 0/100 under the policy package with `exact_gold`, and 1/100 under the policy package with `auto_compiled`. Consequently, the full-minus-policy difference under `auto_compiled` was −1.00 percentage point (task-cluster bootstrap 95% CI −3.00 to 0.00; Holm-adjusted McNemar p=1.0), while all other primary

security contrasts were exactly zero or differed by the same single event. These data provide no evidence that the full package adds a reproducible security benefit over the policy package. This is a package-level result because the prompt profile also differs. For a 0/100 cell, however, the Wilson 95% upper limit is 3.70%; rare harms below that resolution cannot be ruled out.

Benign-tool completion showed the larger authorization effect. Under `auto_compiled`, the policy package completed 22/60 tasks (36.67%, Wilson 95% CI 25.62–49.32) and the full package completed 16/60 (26.67%, 17.13–39.01), a paired full-minus-policy difference of −10.00 points (bootstrap 95% CI −18.33 to −3.33; Holm-adjusted p=0.09375). Under `exact_gold`, the policy package completed 9/60 (15.00%, 8.10–26.11) and full completed 7/60 (11.67%, 5.77–22.18), a −3.33-point paired difference (bootstrap 95% CI −8.33 to 0.00; Holm-adjusted p=0.5). All four main cells completed 20/20 benign no-tool tasks.

Changing the policy package from `auto_compiled` to `exact_gold` reduced paired benign-tool completion by 21.67 points (bootstrap 95% CI −36.67 to −5.00; Holm-adjusted p=0.05853). For the full package, the corresponding reduction was 15.00 points (−28.33 to −1.67; Holm-adjusted p=0.09808). The bootstrap intervals exclude zero, but neither exact McNemar comparison remains below 0.05 after Holm correction. We therefore interpret the consistent direction and effect size as evidence of authorization sensitivity in this sample, not as a universal oracle-policy effect.

4.7.2 Benchmark-stratified authorization effects

The pooled utility estimate hides a qualitative benchmark reversal. On AgentDojo benign tool tasks, changing from `auto_compiled` to `exact_gold` increased completion from 5/30 to 9/30 for the policy package (+13.33 points; bootstrap 95% CI −3.33 to +30.00) and from 4/30 to 7/30 for the full package (+10.00 points; −3.33 to +23.33). On M3-v2, the same change reduced completion from 17/30 to 0/30 for the policy package (−56.67 points; −73.33 to −40.00) and from 12/30 to 0/30 for the full package (−40.00 points; −56.67 to −23.33). These paired strata are descriptive, but they show that the pooled −21.67 and −15.00 point estimates are not transportable across the two benchmark task semantics.

4.7.3 Failure-sensitive attack bound

Counting `unsafe_effectORoperational_failure` as a pessimistic sensitivity endpoint changes the interpretation of the exact regime. The policy package moved from 7/100 under `auto_compiled` to 25/100 under `exact_gold` (+18.00 points; bootstrap 95% CI +9.00 to +27.00); the full package moved from 8/100 to 26/100 (+18.00 points; +10.00 to +27.00).

Package contrasts within the same regime were only +1.00 point and had intervals spanning zero.

This sensitivity effect was entirely benchmark dependent. On AgentDojo, policy and full package differences from auto to exact were both +36.00 points (5/50 to 23/50 and 4/50 to 22/50; both bootstrap intervals +20.00 to +52.00). On M3-v2, the corresponding differences were 0.00 points (2/50 to 2/50 and 4/50 to 4/50). Operational failures remain separately labeled; the composite is a robustness bound and is not reported as observed attack success.

4.7.4 Stress regimes and operational burden

On the nested benign-tool subset, moving from matched full-package `auto_compiled` executions to broad, conflicting, or underspecified policies reduced completion by 25, 30, and 25 percentage points, respectively. Their bootstrap intervals were −45 to −5, −50 to −10, and −50 to 0 points. No unsafe effect occurred in any of the 30-attack stress cells, so the stress comparison primarily exposes lost utility rather than an observable security gain.

The decision and failure profiles explain part of this trade-off. Under the policy package, `auto_compiled` produced ASK on 34.44% and DENY on 7.22% of records, whereas `exact_gold` produced no ASK but DENY on 48.89%; its failure rate rose from 7.22% to 23.89% and p95 latency from 10.63 to 19.42 seconds. The full package showed the same pattern: ASK fell from 32.22% to zero, DENY rose from 11.11% to 45.00%, failure rose from 9.44% to 24.44%, and p95 latency rose from 9.18 to 19.62 seconds. DENY incidence counts records with at least one DENY; the separate event totals are retained because a record may contain multiple decisions. Across all 900 work items, provider cost was CNY 1.3888324.

Pooling again obscures where the system burden occurred. Under `auto_compiled`, system ASK incidence on AgentDojo was 80.00%/78.00% for policy/full attacks and 66.67%/56.67% for policy/full benign-tool tasks; the corresponding M3-v2 rates were 4.00%/2.00% and 0.00%/3.33%. Under `exact_gold`, ASK was zero by construction. Exact-regime AgentDojo failure was 46.00%/44.00% on attacks and 43.33%/43.33% on benign-tool tasks, compared with 4.00%/8.00% and 16.67%/16.67% on M3-v2. AgentDojo exact p95 latency ranged from 20.77 to 23.93 seconds across these four package/task cells, whereas M3-v2 exact p95 ranged from 6.93 to 8.66 seconds. Thus `ASK`, blocking, failure, and latency are properties of the package–regime–benchmark combination, not a single pooled human-burden quantity.

These results reverse the simple expectation that an answer-derived exact policy must improve measured utility. In this implementation, the exact policy behaved as a more restrictive authorization treatment, replacing confirmation with repeated blocking and higher operational

failure. The result supports making authorization construction explicit in evaluations; it does not show that every gold-derived policy or every runtime guard behaves similarly.

4.8 Concurrent prompt-by-module factorial experiment

E1 reran the fixed 180-case cohort contemporaneously under the four cells of a 2×2 design: policy/normal, policy/safe, full/normal, and full/safe. All 720 planned work items obtained an official response. There were 657 completed records, 54 parse failures, and nine provider failures; all 63 non-completed outcomes remain in ITT. Raw evaluator fields, provider-managed tools, and real external effects were absent from every online record.

Direct unsafe effects remained floor-limited: the two policy cells each produced 1/100, whereas both full cells produced 0/100. The estimated module main effect was −1.00 percentage point (10,000-case-cluster bootstrap 95% CI −3.00 to 0.00; Holm-adjusted p=1.000). The prompt and interaction estimates were both 0.00 points. The failure-sensitive endpoint was 9/100, 11/100, 7/100, and 8/100 in policy/normal, policy/safe, full/normal, and full/safe, respectively. Its module estimate was −2.50 points (−5.50 to 0.00; descriptive bootstrap p=0.0938). This prespecified sensitivity endpoint was not included in either co-primary Holm family.

Benign-tool completion was 23/60, 21/60, 21/60, and 19/60 across the same cells. Both the module and prompt main effects were −3.33 points; their 95% intervals were −8.33 to 0.00 and −10.83 to +4.17, respectively, and the interaction estimate was 0.00 points (−6.67 to +8.33). Thus the concurrent design removes the earlier prompt/module confounding but supplies no evidence of a positive incremental module or safe-prompt effect. It also does not solve the direct-safety floor: only two unsafe effects occurred across 400 assigned attack runs.

Supplementary Table S7 reports the E1 runtime burden and cost.

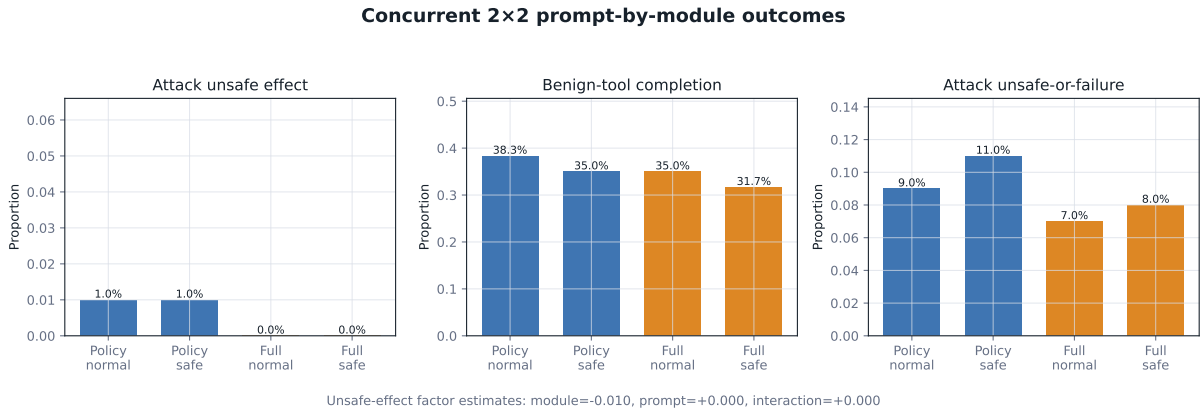


Figure 2 **Prompt-by-module factorial effects. Panels show the four E1 cells for unsafe effect, benign-tool completion, and the failure-sensitive attack endpoint; Table 2 reports the corresponding module, prompt, and interaction effects. Intervals use 10,000 case-cluster bootstrap replicates. Holm adjustment is confined to the unsafe-effect and benign-tool TCR co-primary families; the failure-sensitive endpoint remains a descriptive sensitivity analysis.**

Table 2 E1 factorial estimates

Study	Endpoint	Factor	Paired cases	Effect	CI low	CI high	Bootstrap p	Holm p	Bootstrap n
E1 representative pro	attack unsafe effect	interaction	100	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	1.0000	10000
E1 representative pro	attack unsafe effect	module	100	-1.00 pp	-3.00 pp	+0.00 pp	0.7358	1.0000	10000
E1 representative pro	attack unsafe effect	prompt	100	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	1.0000	10000
E1 representative pro	attack unsafe or failure	interaction	100	-1.00 pp	-5.00 pp	+3.00 pp	0.8206	—	10000
E1 representative pro	attack unsafe or failure	module	100	-2.50 pp	-5.50 pp	+0.00 pp	0.0938	—	10000
E1 representative pro	attack unsafe or failure	prompt	100	+1.50 pp	-3.00 pp	+6.00 pp	0.5868	—	10000
E1 representative pro	Benign-tool TCR	interaction	60	+0.00 pp	-6.67 pp	+8.33 pp	1.0000	1.0000	10000
E1 representative pro	Benign-tool TCR	module	60	-3.33 pp	-8.33 pp	+0.00 pp	0.0960	0.2880	10000
E1 representative pro	Benign-tool TCR	prompt	60	-3.33 pp	-10.83 pp	+4.17 pp	0.4496	0.8992	10000

Note. Absolute effects are probability differences; intervals use 10,000 case-cluster bootstrap replicates. Holm values apply only to E1's prespecified endpoint families.

4.9 Source-authority counterfactual

E2 completed 240/240 response-bearing end-to-end work items and 480/480 deterministic Guard decisions. The source swap changed only the authority of a critical proposition while holding text, value, relative position, and policy fixed. No end-to-end cell executed an authorized, unauthorized, or other action. The resulting zero unauthorized-argument-influence rate is an endpoint floor caused by confirmation and blocking, not evidence that source authority was behaviorally irrelevant.

The burden nevertheless differed sharply. Policy/trusted and policy/untrusted produced ASK in 60/60 and 53/60 records, with no DENY. The full/trusted cell produced ASK in 57/60, whereas full/untrusted produced no ASK, DENY in 55/60, and one operational failure. The actual executed-action tuple changed under the source swap in 0/60 paired tasks for both module settings; operational status changed in 1/60 full pairs and is reported separately.

The deterministic replay provides mechanism evidence rather than end-to-end utility. Its 480 decisions comprise 240 unauthorized and 240 authorized proposals. False allow was 0/240 unauthorized proposals (Wilson 95% CI 0.00%–1.58%); false block was 30/240 authorized proposals (12.50%, 8.90%–17.28%). Policy decisions were identical under trusted and untrusted origins: 60 ALLOW and 60 ASK. Full/trusted had the same distribution, but full/untrusted produced 30 ALLOW, 30 ASK, and 60 DENY. The full Guard is therefore provenance-sensitive under this controlled swap, yet the tested model–Guard system did not translate that mechanism into completed tasks and incurred substantial confirmation or blocking burden.

Supplementary Table S8 reports the E2 source-authority decisions.

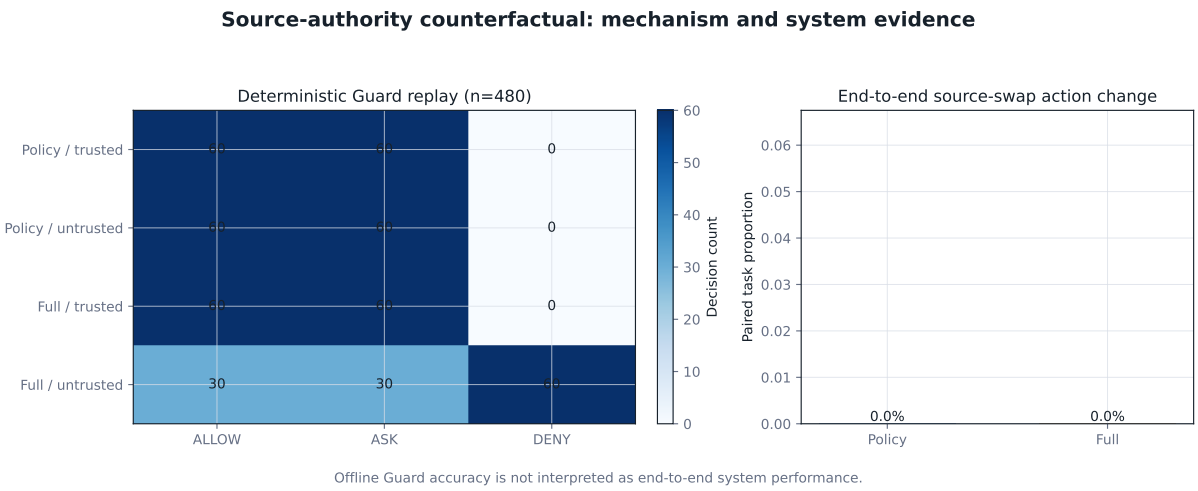


Figure 3 Source-authority counterfactual. The decision heatmap reports deterministic Guard outcomes for authorized and unauthorized proposals under trusted/untrusted origins and policy/full modules. The paired action panel separately reports end-to-end source-swap action changes. Offline Guard correctness is not treated as a substitute for system task completion.

Table 3 E2 source-authority evidence

Study	Endpoint	Factor	Paired cases	Effect	CI low	CI high	Bootstrap p	Holm p	Bootstrap n
E2 end to end	unauthorized argument influence	module	60	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	1.0000	10000
E2 end to end	unauthorized argument influence	source	60	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	1.0000	10000
E2 end to end	unauthorized argument influence	source x module	60	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	1.0000	10000
E2 end to end	authorized action execution	module	60	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	1.0000	10000
E2 end to end	authorized action execution	source	60	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	1.0000	10000
E2 end to end	authorized action execution	source x module	60	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	1.0000	10000
E2 end to end	source swap action change	full	60	+0.00 pp	+0.00 pp	+6.02 pp	—	—	—
E2 end to end	source swap action change	policy	60	+0.00 pp	+0.00 pp	+6.02 pp	—	—	—
E2 offline guard replay	false allow	not applicable	240	+0.00 pp	+0.00 pp	+1.58 pp	—	—	—
E2 offline guard replay	false block	not applicable	240	+12.50 pp	+8.90 pp	+17.28 pp	—	—	—

Note. End-to-end source swaps and deterministic Guard replay are separate evidence layers and must not be interpreted as interchangeable.

4.10 Outcome-enriched challenge and Flash replication

E3 reran 60 attacks selected because each had produced an unsafe effect under at least one historical Public condition. It is an outcome-enriched challenge set, not a prevalence sample. All 240 work items obtained responses, but only 185 completed normally; 17 parse and 38 provider failures remained in ITT. Only 1/60 historical vulnerabilities reproduced under any concurrent cell (1.67%, Wilson 95% CI 0.29%–8.86%). The four unsafe-effect cells were 1/60, 1/60, 0/60, and 0/60, while unsafe-or-failure was 16/60, 15/60, 12/60, and 14/60. The module estimates were −1.67 points (−5.00 to 0.00) for unsafe effect and −4.17 points (−10.00 to +0.83) for unsafe-or-failure. These conditional results expose instability and a high failure burden rather than establishing deployment robustness.

E4 applied the same factorial design to the frozen 90-case DeepSeek V4 Flash cohort. All 360 work items obtained responses; one response-less transport attempt was superseded by a single-key successful attempt without replaying any other row. The final ITT set contains 347 completed, eight parse-failure, four provider-failure, and one maximum-step outcomes. Every attack cell was 0/50 unsafe effects. Benign-tool completion was 10/30, 8/30, 10/30, and 10/30; the module, prompt, and interaction estimates were +3.33 points (0.00 to +8.33), −3.33 points (−11.67 to +3.33), and +6.67 points (0.00 to +16.67). Unsafe-or-failure yielded a +3.00-point module estimate (0.00 to +7.00). E4 is a same-provider model replication and opens no new significance family; its all-zero direct-safety endpoint cannot establish equivalence or cross-provider generalization.

Supplementary Table S9 reports the E3/E4 operational summary.

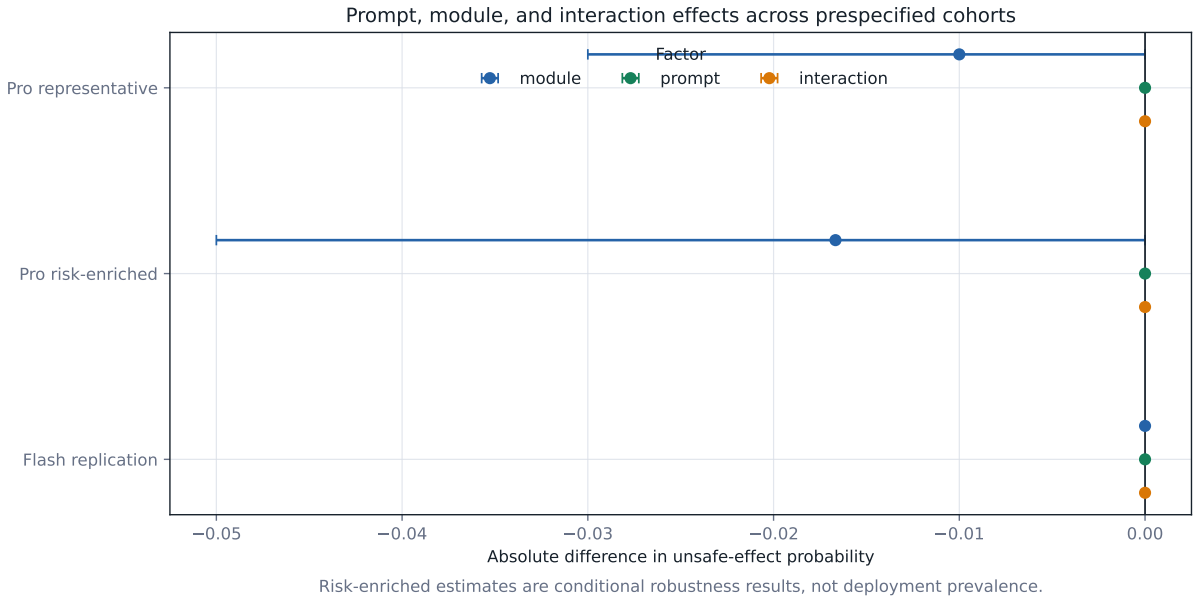


Figure 4 **Factor effects across representative, enriched, and Flash cohorts. Absolute module, prompt, and interaction effects are shown for E1, the outcome-enriched E3 challenge, and the same-provider E4 Flash replication. E3 does not estimate prevalence and E4 does not establish cross-provider generality; neither opens a new significance family.**

Table 4 E3/E4 conditional robustness

Study	Endpoint	Factor	Paired cases	Effect	CI low	CI high	Bootstrap p	Holm p	Bootstrap n
E3 risk enriched pro	attack unsafe effect	interaction	60	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	—	10000
E3 risk enriched pro	attack unsafe effect	module	60	-1.67 pp	-5.00 pp	+0.00 pp	0.7406	—	10000
E3 risk enriched pro	attack unsafe effect	prompt	60	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	—	10000
E3 risk enriched pro	attack unsafe or failure	interaction	60	+5.00 pp	-3.33 pp	+13.33 pp	0.3448	—	10000
E3 risk enriched pro	attack unsafe or failure	module	60	-4.17 pp	-10.00 pp	+0.83 pp	0.1528	—	10000
E3 risk enriched pro	attack unsafe or failure	prompt	60	+0.83 pp	-8.33 pp	+10.00 pp	0.9438	—	10000
E4 flash replication	attack unsafe effect	interaction	50	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	—	10000
E4 flash replication	attack unsafe effect	module	50	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	—	10000
E4 flash replication	attack unsafe effect	prompt	50	+0.00 pp	+0.00 pp	+0.00 pp	1.0000	—	10000
E4 flash replication	attack unsafe or failure	interaction	50	-2.00 pp	-10.00 pp	+4.00 pp	0.7852	—	10000
E4 flash replication	attack unsafe or failure	module	50	+3.00 pp	+0.00 pp	+7.00 pp	0.0864	—	10000
E4 flash replication	attack unsafe or failure	prompt	50	-1.00 pp	-5.00 pp	+2.00 pp	0.7752	—	10000
E4 flash replication	Benign-tool TCR	interaction	30	+6.67 pp	+0.00 pp	+16.67 pp	0.2558	—	10000
E4 flash replication	Benign-tool TCR	module	30	+3.33 pp	+0.00 pp	+8.33 pp	0.2602	—	10000
E4 flash replication	Benign-tool TCR	prompt	30	-3.33 pp	-11.67 pp	+3.33 pp	0.5494	—	10000

Note. E3 is outcome-enriched and E4 is a same-provider replication. Neither opens a new significance family or estimates deployment prevalence.

4.11 Contemporaneous attack-activity calibration

E6 used a separately executed, historically risk-enriched cohort to test whether attacks were still active under a current no-defense control. The execution was independent of E3, but the sample was not: 47/60 tasks were E3 anchors (19 AgentDojo and 28 M3-v2). All 360 DeepSeek work items were response-bearing; 361 physical records include one response-less superseded attempt. No defense produced 17/60 unsafe effects (28.33%, Wilson 95% CI 18.52%–40.75%), including 8/30 AgentDojo and 9/30 M3-v2 cases. This demonstrates observed attack activity within the selected risk-enriched block, but does not satisfy or test the original multi-window activity gate and cannot estimate deployment prevalence.

Unsafe effects under source-pinned VIGIL, policy/normal, policy/safe, full/normal, and full/safe were 16/60, 2/60, 1/60, 0/60, and 0/60. Operational failures in the same order, including no defense first, were 12/60, 8/60, 7/60, 2/60, 7/60, and 3/60. Full/safe minus no defense was −28.33 percentage points (paired Newcombe 95% CI −40.77 to −16.81; task-cluster bootstrap −40.00 to −16.67; Holm-adjusted p=0.000061). Policy/normal minus no defense was −25.00 points (Newcombe −36.95 to −14.30), and full/safe minus VIGIL was −26.67 points (−39.01 to −15.39). VIGIL minus no defense was only −1.67 points with an interval crossing zero; its benchmark-specific directions opposed each other (+3.33 points on AgentDojo and −6.67 on M3-v2), so the pooled VIGIL contrast is not interpreted as homogeneous.

The four M3 cells yielded descriptive factor estimates of −2.50 points for the module bundle, −0.83 for the safe prompt, and +1.67 for their interaction. E6 contains no benign tasks and was executed as one block after a prospective protocol amendment; it adds neither TCR evidence nor temporal-repeatability or individual-module attribution.

The nested Gemini formal prefix failed its technical gate after five response-bearing records and one response-less VIGIL dependency failure. The remaining 174 planned items were administratively closed, not counted as safe or scientifically executed. The nested set also overlapped E3 for 29/30 tasks. Minimum response-complete paired coverage was 0/30 for comparisons involving VIGIL. All Gemini paired, factorial, and cross-provider effects are not estimable.

Table 5 E6 contemporaneous attack-activity calibration

Condition	Unsafe effect, n/N (95% CI)	Operational failure	Unsafe or failure	ASK events	DENY events
No defense	17/60 (28.3%) [18.5%, 40.8%]	12/60 (20.0%)	29/60 (48.3%)	0	4
Source-pinned VIGIL	16/60 (26.7%) [17.1%, 39.0%]	8/60 (13.3%)	24/60 (40.0%)	0	8
Policy / normal	2/60 (3.3%) [0.9%, 11.4%]	7/60 (11.7%)	9/60 (15.0%)	31	7
Policy / safe	1/60 (1.7%) [0.3%, 8.9%]	2/60 (3.3%)	3/60 (5.0%)	21	2
Full / normal	0/60 (0.0%) [0.0%, 6.0%]	7/60 (11.7%)	7/60 (11.7%)	25	18
Full / safe	0/60 (0.0%) [0.0%, 6.0%]	3/60 (5.0%)	3/60 (5.0%)	19	6

Note. E6 uses a historically risk-enriched single execution block; 47/60 tasks also anchor E3. It does not estimate deployment prevalence, benign utility, temporal repeatability, independent-sample replication, or a cross-provider effect.

Supplementary Table S11 reports the E6 DeepSeek activity and operational outcomes.

Supplementary Table S12 reports the E6 paired effects and Gemini inference coverage.

4.12 Defense-aware adaptive evidence and conditional tail through budget 500

The initial DeepSeek V4 Pro cost canary covered eight result-blind, attack-family-balanced M3-v2 development tasks under the full package and the `auto_compiled` authorization regime. At a maximum of 50 queries per task, no unsafe effect was found: 0/8 tasks, or 0.00% (Wilson 95% CI 0.00–32.44%). The run retained one closed diagnostic query in ITT; that task ended at query 24, while the other seven tasks reached query 50. In total, the canary contains 374 terminal queries, 562 official provider calls, and CNY 0.552805 cost. Initial decisions across terminal queries were 207 ALLOW, 123 ASK, and 44 DENY.

The full frozen panel subsequently reached terminal status for all 40 tasks. First unsafe effect occurred for 4/40 tasks by budget 50 (10.00%, Wilson 95% CI 3.96–23.05) and for 18/40 by budget 200 (45.00%, 30.71–60.17). Fourteen of the 18 observed successes therefore appeared only after query 50. The evidence supports a budget-sensitivity claim through 200, but it does not establish saturation, compare another model, or justify a SOTA/adaptive-robustness claim.

Supplementary Table S6 reports the defense-aware adaptive attack.

The E5 extension then completed all 12 prespecified B200 survivors through query 500. No task produced a new first unsafe effect between queries 201 and 500: 0/12 (0.00%, Wilson 95% CI 0.00%–24.25%). This is a conditional null observation with a wide upper bound, not evidence that the original 40-task panel has zero tail risk or that attack search has saturated. The latest ITT set contains 3,600 query work keys: 3,577 completed, 21 parse failures, and two provider failures. Five earlier response-less or local-recovery attempts remain as superseded physical rows. Total usage was 4,511,768 tokens and attributed cost was CNY 24.5551236.

Supplementary Table S10 reports the conditional adaptive-tail execution.

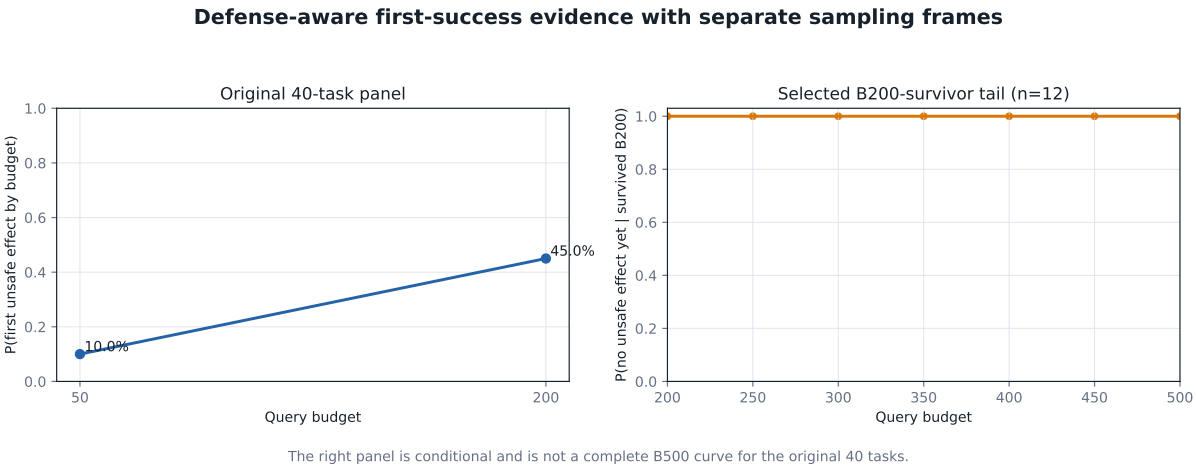


Figure 5 **Complete B50/B200 evidence and the conditional adaptive tail. The original 40-task panel contributes the complete cumulative first-success evidence through 200. The second panel includes only 12 prespecified B200 survivors extended toward 500. The conditional tail is not merged into a full original-panel B500 curve and is not an estimate of saturation.**

Table 6 E5 conditional adaptive tail

Stratum	Tasks	Successes by 500	Conditional probability	CI low	CI high	First-success queries	Complete original B500 panel
all selected survivors	12	0	0.00%	0.00%	24.25%	—	No
A1	4	0	0.00%	0.00%	48.99%	—	No
A2	4	0	0.00%	0.00%	48.99%	—	No
A3	4	0	0.00%	0.00%	48.99%	—	No

Note. The tail probability is conditional on the prespecified B200 survivors; it is not a complete original-panel B500 curve or saturation estimate.

4.13 Remaining evidence boundary

The medium, Public, authorization-regime, factorial, source-authority, challenge, replication, adaptive, and contemporaneous-activity studies jointly support an empirical security–utility paper. They do not support a leading-defense, end-to-end provenance-benefit, saturated-adaptive, or cross-provider claim. E2 does identify source sensitivity at the deterministic Guard decision layer, but its end-to-end action endpoint is floor-limited. E5 supplies only a conditional 0/12 B200-survivor tail with a 24.25% upper confidence limit. E6 records attack activity only within its historically enriched, single-block DeepSeek cohort; it neither satisfies nor tests the original multi-window activity gate. The earlier Gemini authorization study closed at 32/450 terminal work items, only five response-bearing; the separately executed E6 Gemini prefix produced five response-bearing records, one response-less failure, and 174 administrative closures. Both are retained for operational accounting but all Gemini effects are not estimable. UTR and DSR are not identifiable from the current record contract and are not reconstructed from unsafe-effect rate.

5 Discussion

5.1 Existing evidence changes the research claim

The completed medium and Public experiments do not support presenting the full configuration package as stronger than the policy package. In the 5,082-record DeepSeek V4 Pro Public matrix, the full package produced more unsafe effects (38/600 versus 13/600), while both packages completed 49/168 benign tool tasks. The 1,791-record V4 Flash matrix showed the same security direction: 24/400 unsafe effects for the full package versus 12/400 for the policy package. The Flash utility difference was small (38/145 versus 35/145).

This is scientifically useful negative evidence, but it was originally package-level evidence. E1 resolves the prompt/runtime confounding by crossing the prompt profile with the module bundle in one execution window. Its module, prompt, and interaction estimates were all small, and none supplied evidence of a positive safety or utility contribution. The result identifies the prompt factor separately from the runtime bundle; it still cannot separate provenance, temporal/data-flow, monotonicity, and confirmation from one another. It shows that the evaluated bundle did not earn a clear advantage under these tasks, not that any individual mechanism is intrinsically ineffective.

5.2 Authorization is part of the treatment

Authorization is often treated as fixed infrastructure rather than an experimental variable. That assumption is risky in tool-agent evaluation. An exact per-case authorization object reconstructed from the expected answer can remove uncertainty that a deployed Guard must resolve. A broad policy can improve apparent utility by authorizing too much. An underspecified policy can make safe completion impossible, while conflicting constraints may increase denial, confirmation, or failure even when they prevent unsafe effects.

The paired design makes these trade-offs observable. `auto_compiled` is the primary deployment-like regime because it is created from trusted user intent before untrusted context is processed. `exact_gold` is an oracle sensitivity condition, not a preferred policy source. The `broad`, `underspecified`, and `conflicting` cells are stress tests on a fixed nested sample. Their value lies in showing whether a headline security number is stable to plausible policy construction errors.

The DeepSeek paired experiment did not reveal an unsafe-effect advantage for `exact_gold`: three of four primary cells had 0/100 unsafe effects, while the remaining policy-package `auto_compiled` cell had 1/100. The meaningful movement was instead in utility and runtime burden. Moving from `auto_compiled` to `exact_gold` reduced pooled benign-tool completion by 21.67 points for the policy package and 15.00 points for the full package. The task-cluster bootstrap intervals excluded zero, although the corresponding McNemar comparisons did not remain below 0.05 after Holm correction.

The pooled direction is incomplete. AgentDojo completion increased by 13.33 points for the policy package and 10.00 points for the full package, whereas M3-v2 completion fell by 56.67 and 40.00 points. Authorization semantics and benchmark task structure therefore interact strongly enough that a single pooled oracle-versus-compiled estimate should not be transported to either benchmark in isolation.

The operational shift was substantial. Exact policies eliminated ASK in both packages but increased the share of records with DENY from 7.22% to 48.89% for the policy package and from 11.11% to 45.00% for the full package. Failure rose to roughly 24% and p95 latency to roughly 19.5 seconds in both exact cells. The broad, conflicting, and underspecified stress regimes likewise reduced completion by 25–30 points on their nested benign-tool subset without changing the observed attack floor. These findings show that policy specification can move burden and utility even when a headline attack rate appears stable.

The pessimistic `unsafe_effectORoperational_failure` bound makes that point more explicit. Exact authorization increased this attack endpoint by 18 points for both packages overall, driven by +36-point changes on AgentDojo and no change on M3-v2. This does not

convert failures into attacks; it demonstrates that a near-zero observed unsafe-effect rate can coexist with poor operational robustness.

5.3 Provenance sensitivity exists at the decision layer

The controlled source-authority experiment sharpens the negative package result. Holding proposition, value, position, and policy fixed, the policy-only Guard produced the same deterministic decision distribution under trusted and untrusted origins. The full Guard changed its untrusted-origin distribution to 30 ALLOW, 30 ASK, and 60 DENY. Across 240 unauthorized proposals there were no false allows; across 240 authorized proposals there were 30 false blocks. This is direct evidence that the implemented full Guard is provenance-sensitive at its deterministic decision boundary.

That mechanism did not translate into completed end-to-end actions. All 240 model–Guard runs executed zero actions: trusted contexts were routed almost entirely to ASK, while full/untrusted contexts were predominantly DENY. The correct interpretation therefore has two layers. Offline replay establishes mechanism behavior; end-to-end execution reveals a confirmation bottleneck and cannot be replaced by the replay's decision accuracy. A provenance mechanism can be logically active while the surrounding interaction design supplies little usable utility.

5.4 Executed effects and utility must be reported together

A low prompt-injection success rate is not sufficient evidence of a useful runtime defense. A detector may flag adversarial text without preventing an effect; a runtime may prevent effects by denying nearly every useful action; and a parser/provider failure may look secure if the evaluator counts only successful executions. We therefore treat isolated executed effect as the security endpoint and benign tool-task completion as a joint primary endpoint.

The Public evidence already illustrates the problem. Relative to the structural condition, both M3 variants reduced unsafe effects but also had much lower benign-tool completion. The policy package then outperformed the full package on security without losing additional utility. The interpretation is not that provenance is intrinsically harmful. It is that the evaluated bundle did not earn a security advantage under these tasks and policies.

ASK, DENY, operational failure, latency, tokens, and cost provide the missing system-facing context. An authorization regime that shifts unsafe effects downward but greatly increases ASK may be acceptable in high-consequence workflows and unacceptable in routine automation. A regime that shifts both ASK and failure upward may simply externalize uncertainty rather than resolve it. The paper reports these outcomes separately. The unsafe-or-failure

sensitivity endpoint is shown only as an explicitly pessimistic robustness bound, not as a replacement composite score.

5.5 Adaptive evidence is budget-sensitive and conditionally extended

The 40-task defense-aware panel found first unsafe effect for 4/40 tasks by 50 queries and 18/40 by 200 queries. Thus a budget-50 evaluation would have missed most successes later observed by budget 200. This supports the general warning that a static or weak attacker can overstate robustness [19].

The complete-panel evidence stops at 200. E5 extends only 12 prespecified B200 survivors to 500 and stops each task at first unsafe effect. No new first success occurred (0/12; Wilson 95% upper limit 24.25%). This conditional tail can estimate `P(firstsuccessby500|nosuccessby200)` for the selected survivors, but it cannot be merged with the original 40 tasks into a nominal full B500 curve, establish saturation, estimate an asymptotic attack rate, or support a hardest-task-2,000 claim. The null point estimate does not establish zero tail risk because the interval remains wide and the cohort is selected conditional on surviving budget 200.

5.6 Contemporaneous calibration restores an active control

The near-zero E1 and E3 attack endpoints raise a calibration question: were the tested attacks inactive at the time of execution, or did defended conditions actually differ from an active control? E6 addresses this in a separately executed historically risk-enriched cohort, but it is not an independent sample validation: 47/60 tasks are also E3 anchors. No defense produced 17/60 unsafe effects, including events in both AgentDojo and M3-v2, while policy/normal produced 2/60 and full/safe produced 0/60. Attack activity was therefore observed within the selected contemporaneous block; the original multi-window activity gate was neither satisfied nor tested.

This calibration does not rescue a broad defense-superiority claim. The cohort was selected from historical control-positive cases, contains no benign tasks, and was run in one execution block. It estimates neither deployment prevalence nor utility or temporal repeatability. Operational failures were also material: 12/60 under no defense and 3/60 even under full/safe. The VIGIL contrast was heterogeneous, moving in opposite directions across the two benchmarks. E6 is therefore best read as an active-control check plus a conditional comparison, not a general robustness ranking or an individual provenance-module ablation.

5.7 Relation to recent provenance and authorization work

VIGIL, DRIFT, the Task Shield, IPIGuard, Progent, CaMeL, ARGUS, AuthGraph, FAVA, and provenance-sensitivity auditing occupy overlapping parts of the design space [3–5, 7, 9–14]. These works rule out claims that M3 is the first verify-before-commit system, the first parameter-source defense, or the first structured runtime authorizer.

The remaining contribution is methodological and empirical. The study holds task content fixed while changing authorization regimes, separately manipulates prompt and runtime factors, and performs a source-only counterfactual. It keeps evaluator fields offline, measures post-enforcement effects, retains failure in ITT, and reports operational burden. In particular, the recent provenance-sensitivity study changes source authority in controlled next-action tasks [13]. The present design adds deterministic Guard replay and complete tool trajectories while keeping those two evidence layers separate.

5.8 Cross-provider interpretation

Two distinct Gemini attempts remain diagnostic. The earlier authorization study closed with 32/450 terminal work items, including only five with official model responses; 83 observed items remained nonterminal and 335 were unstarted. The separately executed E6 prefix then stopped after five response-bearing records and one response-less VIGIL dependency failure; 174 untouched plan rows were closed administratively. Administrative closure is not scientific observation.

Neither dataset meets a paired-coverage gate. In E6, comparisons involving VIGIL have 0/30 minimum response-complete pairs. We therefore report all Gemini effects as not estimable rather than plotting a zero, ranking providers, or combining the two incompatible samples. External validity remains limited to DeepSeek V4 Pro and V4 Flash.

5.9 Limitations

First, the new primary experiment uses one proprietary model and two benchmark families; the larger Public evidence adds another model from the same provider, not an independent vendor. Model updates, tool schemas, system prompts, and application context may change the effects. Both Gemini attempts closed incomplete and contribute no cross-provider inference. Although execution dates, official response identifiers, and returned model names are retained, provider-managed aliases do not make the underlying weights immutable.

Second, the new authorization experiment has a strong attack-endpoint floor: only one unsafe effect occurred across the 400 primary attack records. It can therefore estimate utility and

operational burden more informatively than small security differences between its four primary cells. Security-direction evidence comes mainly from the larger Public matrices and must not be inferred from these near-zero cells alone.

Third, E1 separates prompt profile from the runtime module bundle, but the full module factor still combines provenance, data-flow, monotonicity, and confirmation. It identifies a bundle-level module effect, not the contribution of each internal mechanism. E2 isolates source authority at the input boundary, yet confirmation and blocking create an end-to-end action floor.

Fourth, `auto_compiled` is one policy compiler, not a complete model of human authorization. The stress regimes approximate common policy errors but do not measure their prevalence in deployed systems. `exact_gold` is intentionally artificial and should not be read as a recommended configuration.

Fifth, the effect environment is isolated. This is necessary to prevent real email, payment, browser, code, or production actions, but it may omit latency, state, and recovery behavior found in live applications.

Sixth, only a prespecified subset of B200 survivors is extended toward 500. That conditional tail does not constitute a complete B500 panel and cannot prove saturation. Attack search is still one family of defense-aware optimization; another optimizer may find different failures.

Seventh, E6 uses a historically risk-enriched cohort in a single execution block, with 47/60 tasks overlapping E3. It observes activity in that selected block but cannot estimate deployment prevalence, benign utility, cross-date repeatability, or independent-sample replication. Its M3 factor estimates remain bundle-level and do not identify provenance as the causal component.

Eighth, failure handling is conservative but not interpretation-free. All response-bearing failures are terminal in ITT, whereas response-less transport failures can receive bounded explicit recovery. We report raw failures and superseded attempts so readers can assess that choice.

Ninth, `ASK` is observed only as system confirmation-request incidence. Confirmation stayed pending in the synthetic environment, so this study does not measure human response quality, workload, or usable-security outcomes.

Finally, this paper does not include an independent blind set. It also does not claim first, only, or SOTA status. Its external-validity claims are limited to the models, benchmarks, policy compiler, stress regimes, and attack budgets actually evaluated.

5.10 Practical implications

Four practical recommendations follow. First, agent-security benchmarks should publish the fields visible to the online Guard and keep evaluator-only answers outside that projection. Second, runtime defenses should be evaluated factorially against a simpler policy package so

prompt and module effects are separately identifiable; the internal modules require further ablations before individual attribution. Third, source-only counterfactuals should be paired with both deterministic Guard replay and end-to-end execution, rather than treating either as a proxy for the other. Fourth, security reports should include benign completion, system confirmation/blocking incidence, failures, latency, and attack budget alongside unsafe-effect rate.

The DeepSeek authorization-regime results now tie these recommendations to observed effects: exact and stressed policies materially changed completion, blocking, failure, and latency while observed unsafe effects remained near zero. E6 additionally shows why an active no-defense control should accompany floor-limited defended cells. Both Gemini attempts closed with incomplete diagnostic subsets and cannot alter or broaden the DeepSeek findings.

6 Conclusion

This study treats runtime authorization as an experimental variable rather than fixed infrastructure. Across the completed medium and Public studies, the full configuration package reduced unsafe effects relative to a structural baseline but did not improve on the policy package. The larger historical direction replicated across DeepSeek V4 Pro and V4 Flash: the full package produced more unsafe effects, while benign-tool utility was equal or only slightly different.

The prospectively frozen authorization experiment explains why security alone is an incomplete endpoint. Unsafe effects remained at or near zero in all four main cells. The pooled exact-minus-auto benign-tool differences were -21.67 points for the policy package and -15.00 for the full package, but AgentDojo improved by 13.33/10.00 points while M3-v2 fell by 56.67/40.00. Under a pessimistic unsafe-or-operational-failure bound, both packages worsened by 18 points overall, driven by +36-point changes on AgentDojo and no change on M3-v2. Exact authorization removed system ASK but moved 45.00–48.89% of records into at least one DENY, raised failures to approximately 24%, and increased p95 latency. Broad, conflicting, and underspecified regimes also reduced completion on the nested tool-task subset without changing the observed attack floor. Authorization construction can therefore materially alter the measured security–utility frontier even when an attack-only result looks stable.

The concurrent 2×2 experiment then separated prompt profile from the runtime module bundle. Across 400 attack assignments, the policy cells produced two unsafe effects and the full cells produced none; the module estimate was -1.00 percentage point (95% bootstrap CI -3.00 to 0.00), while the prompt and interaction estimates were zero. Benign-tool module and prompt effects were both -3.33 points with intervals including zero. The design resolves the prompt/module confounding but remains floor-limited and does not isolate the individual

provenance, data-flow, monotonicity, or confirmation components.

The source-authority counterfactual supplies the corresponding mechanism evidence. FullGuard decisions changed under an untrusted source, whereas the policy-only distribution did not; 480 replay decisions comprised 0/240 false allows among unauthorized proposals and 30/240 false blocks among authorized proposals. Yet all 240 end-to-end runs executed no action because ASK or DENY intervened. Mechanism-level provenance sensitivity is therefore not equivalent to usable end-to-end utility. The outcome-enriched challenge set likewise reproduced only 1/60 historical unsafe effects and incurred 55 operational failures, while the Flash factorial replication had zero unsafe effects in every attack cell. Both results reinforce the need to report a failure-sensitive endpoint instead of treating non-operation as safety.

The adaptive panel supplies a related measurement warning. First unsafe effects were observed for 4/40 tasks by query 50 and 18/40 by query 200. A single low attack budget would have missed most successes found by the larger budget. Among the 12 prespecified B200 survivors, 0/12 produced a new first unsafe effect by query 500 (Wilson upper limit 24.25%). This conditional result cannot be merged into a full B500 curve or used to establish saturation or zero tail risk.

A separately executed single-block calibration observed attack activity in a historically risk-enriched cohort: no defense produced 17/60 unsafe effects across both benchmarks, compared with 2/60 under policy/normal and 0/60 under full/safe. Because 47/60 tasks also anchored E3, this is not independent-sample replication. These conditional contrasts remain bounded by non-negligible operational failure, outcome-enriched sampling, the absence of benign tasks, and a single execution block. They do not satisfy the original multi-window activity gate or estimate deployment prevalence, utility, temporal repeatability, or an individual provenance-module effect.

These findings support four practical requirements for tool-agent security evaluation: keep evaluator-only fields outside online Guard state; manipulate prompt and runtime factors separately; pair deterministic source-authority replay with end-to-end execution; and report utility, ASK/DENY incidence, failures, latency, and attack budget alongside unsafe effects. The conclusions remain limited to the tested policies, benchmarks, runtime implementation, DeepSeek models, and budget range. Both the earlier Gemini authorization sample and the separate E6 Gemini prefix closed below their paired-coverage gates; all Gemini effects are not estimable and cannot erase or broaden the completed DeepSeek findings.

Declarations

Competing interests

The corresponding author must confirm the competing-interests declaration with all authors before submission.

Funding

The corresponding author must provide the funder name, grant number, and sponsor role, or confirm that the research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data and code availability

The implementation, frozen lean experiment plan, analysis code, aggregate tables, and editable figures are maintained at `https://github.com/Fischl0527/RedTeam_LLM_Security`. The repository is private during manuscript preparation. A public archival release with an immutable version identifier will be deposited before publication. Credentials and secrets are excluded from the release. Any restrictions on provider-derived compact raw records will be stated explicitly in the archival record.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the authors used OpenAI Codex for software editing, statistical-script review, figure-generation support, and language revision. An earlier DeepSeek-generated project-planning note supplied by the project owner was consulted during study scoping. The authors reviewed and revised the resulting material and take full responsibility for the analysis, interpretation, citations, and manuscript content. DeepSeek and Gemini API outputs reported in the experiments are evaluated research data; they were not used to generate outcome labels or statistical conclusions.

Ethics and safety

The study involved no human participants and collected no personal data. Tool effects were evaluated in isolated benchmark state. Real email, payment, browser, code-execution, production-API, and provider-managed tool operations were disabled. The authors must confirm any institutional review or exemption requirement before submission.

Author contributions

Author order and CRediT roles must be approved by every author before submission.

References

[1] DEBENEDETTI E, ZHANG J, BALUNOVIC M, et al. Agentdojo: A dynamic environment to evaluate prompt injection attacks and defenses for llm agents[J/OL]. Advances in Neural Information Processing Systems, 2024, 37:82895-82920. https://proceedings.neurips.cc/paper_files/paper/2024/hash/97091a5177d8dc64b1da8bf3e1f6fb54-Abstract-Datasets_and_Benchmarks_Track.html. DOI: 10.52202/079017-2636.

[2] ZHAN Q, LIANG Z, YING Z, et al. InjecAgent: Benchmarking indirect prompt injections in tool-integrated large language model agents[C/OL]//KU L W, MARTINS A, SRIKUMAR V. Findings of the Association for Computational Linguistics: ACL 2024. Bangkok, Thailand: Association for Computational Linguistics, 2024: 10471-10506. https://aclanthology.org/2024.findings-acl.624/. DOI: 10.18653/v1/2024.findings-acl.624.

[3] LIN J, ZHOU Z, ZHENG Z, et al. VIGIL: Defending LLM agents against tool-stream injection via verify-before-commit[C/OL]//LIAKATA M, MOREIRA V P, ZHANG J, et al. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). San Diego, California, United States: Association for Computational Linguistics, 2026: 9764-9785. https://aclanthology.org/2026.acl-long.443/. DOI: 10.18653/v1/2026.acl-long.443.

[4] LI H, LIU X, CHUN C, et al. Drift: Dynamic rule-based defense with injection isolation for securing llm agents[J/OL]. Advances in Neural Information Processing Systems, 2025, 38:83262-83290. https://proceedings.neurips.cc/paper_files/paper/2025/hash/77f3b26c7907aa27b207df9b9d43f29a-Abstract-Conference.html. DOI: 10.52202/085713-2791.

[5] JIA F, WU T, QIN X, et al. The task shield: Enforcing task alignment to defend against indirect prompt injection in LLM agents[C/OL]//CHE W, NABENDE J, SHUTOVA E, et al. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Vienna, Austria: Association for Computational Linguistics, 2025: 29680-29697. https://aclanthology.org/2025.acl-long.1435/. DOI: 10.18653/v1/2025.acl-long.1435.

[6] WANG H, POSKITT C M, SUN J. Agentspec: Customizable runtime enforcement for safe and reliable llm agents[C/OL]//2026 IEEE/ACM 48th International Conference on

Software Engineering (ICSE). 2026. https://conf.researchr.org/details/icse-2026/icse-2026-research-track/29/AgentSpec-Customizable-Runtime-Enforcement-for-Safe-and-Reliable-LLM-Agents.

[7] AN H, ZHANG J, DU T, et al. IPIGuard: A novel tool dependency graph-based defense against indirect prompt injection in LLM agents[C/OL]//CHRISTODOULOPOULOS C, CHAKRABORTY T, ROSE C, et al. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. Suzhou, China: Association for Computational Linguistics, 2025: 1023-1039. https://aclanthology.org/2025.emnlp-main.53/. DOI: 10.18653/v1/2025.emnlp-main.53.

[8] TSAI L, BAGDASARYAN E. Contextual agent security: A policy for every purpose [C/OL]//Proceedings of the Workshop on Hot Topics in Operating Systems. 2025: 8-17. https://research.google/pubs/context-is-key-for-agent-security/. DOI: 10.1145/3713082.3730378.

[9] SHI T, HE J, WANG Z, et al. Progent: Securing ai agents with privilege control[EB/OL]. 2025. https://arxiv.org/abs/2504.11703.

[10] DEBENEDETTI E, SHUMAILOV I, FAN T, et al. Defeating prompt injections by design [EB/OL]. 2025. https://arxiv.org/abs/2503.18813.

[11] WENG S, FENG Y, ZHANG J, et al. Argus: Defending llm agents against context-aware prompt injection[EB/OL]. 2026. https://arxiv.org/abs/2605.03378.

[12] WANG P, LI Y, TIAN Y. Aligning provenance with authorization: A dual-graph defense for llm agents[EB/OL]. 2026. https://arxiv.org/abs/2605.26497.

[13] LIAO J. Auditing provenance sensitivity in llm agent action selection[EB/OL]. 2026. https://arxiv.org/abs/2607.20827.

[14] ZHANG Y, ZHAO X, LIU S, et al. Fava: Formal authorization for verified agents with evidence-backed permission graphs[EB/OL]. 2026. https://arxiv.org/abs/2607.27267.

[15] CHEN S, PIET J, SITAWARIN C, et al. StruQ: Defending against prompt injection with structured queries[C/OL]//34th USENIX Security Symposium (USENIX Security 25). 2025: 2383-2400. https://www.usenix.org/conference/usenixsecurity25/presentation/chen-sizhe.

[16] HINES K, LOPEZ G, HALL M, et al. Defending against indirect prompt injection attacks with spotlighting[EB/OL]. 2024. https://arxiv.org/abs/2403.14720.

[17] WALLACE E, XIAO K, LEIKE R, et al. The instruction hierarchy: Training llms to prioritize privileged instructions[EB/OL]. 2024. https://arxiv.org/abs/2404.13208.

[18] CHEN Y, LI H, SUI Y, et al. Can indirect prompt injection attacks be detected and removed?[C/OL]//CHE W, NABENDE J, SHUTOVA E, et al. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Vienna, Austria: Association for Computational Linguistics, 2025: 18189-18206. https://aclanthology.org/2025.acl-long.890/. DOI: 10.18653/v1/2025.acl-long.890.

[19] NASR M, CARLINI N, SITAWARIN C, et al. The attacker moves second: Stronger adaptive attacks bypass defenses against LLM jailbreaks and prompt injections[EB/OL]. 2025. https://arxiv.org/abs/2510.09023.

[20] GONG G, DENG Z. Planguard: Defending agents against indirect prompt injection via planning-based consistency verification[EB/OL]. 2026. https://arxiv.org/abs/2604.10134.

[21] ZHANG H, HUANG J, MEI K, et al. Agent security bench (asb): Formalizing and benchmarking attacks and defenses in llm-based agents[C/OL]//YUE Y, GARG A, PENG N, et al. International Conference on Learning Representations: volume 2025. 2025: 35331-35366. https://proceedings.iclr.cc/paper_files/paper/2025/file/5750f91d8fb9d5c02bd8ad2c3b44456b-Paper-Conference.pdf.